

Executive Summary

The Economic Viability of BESS for North-West Transmission Constraint Relief

The assets the grid physically prefers (4h–8h) face the highest economic hurdles, while the assets most readily funded by current markets (1h–2h) provide less effective relief for persistent congestion.

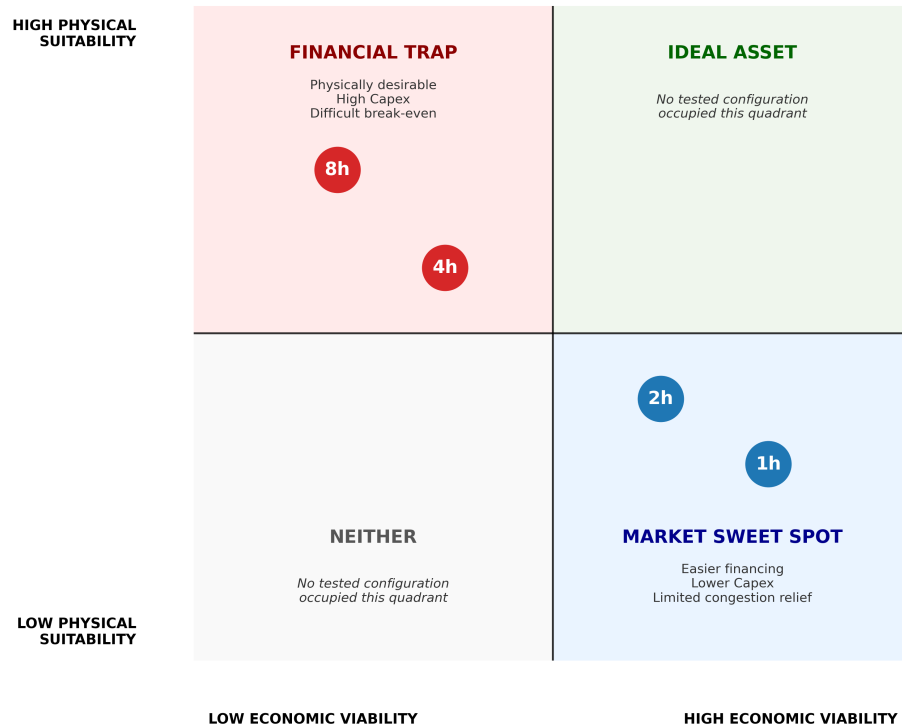


Figure 1 — The Physical-Economic Tension.

The Core Insight

This study evaluates whether persistent North-West transmission constraints create economically viable opportunities for standalone Battery Energy Storage Systems (BESS). By bridging physical grid behaviour with rigorous project finance modelling, the analysis reveals a fundamental market failure: **the GB flexibility market is trapped in a “pincer movement.”**

Constraint-management revenue alone is insufficient to justify standalone investment. More critically, **constraint revenue cannot rescue a structurally impaired merchant case.** Even when tested as a marginal top-up to a stacked revenue strategy, the required break-even performance from other markets (£100k–£380k/MW-year) is materially above recent market norms

that dominate the connection queue, and commercially unachievable for long-duration assets.

Consequently, the free market is currently incapable of self-correcting to deploy the physical assets the grid actually needs.

1. The Physical Reality: Friction Dominates Opportunity

Initial screening identified substantial theoretical “Gross Opportunity” (£500m+ over the study period) concentrated within a small number of North-West constraint groups (e.g., GALLEX, NKILGRMO). However, when realistic physical operating limits are applied, the accessible revenue collapses.

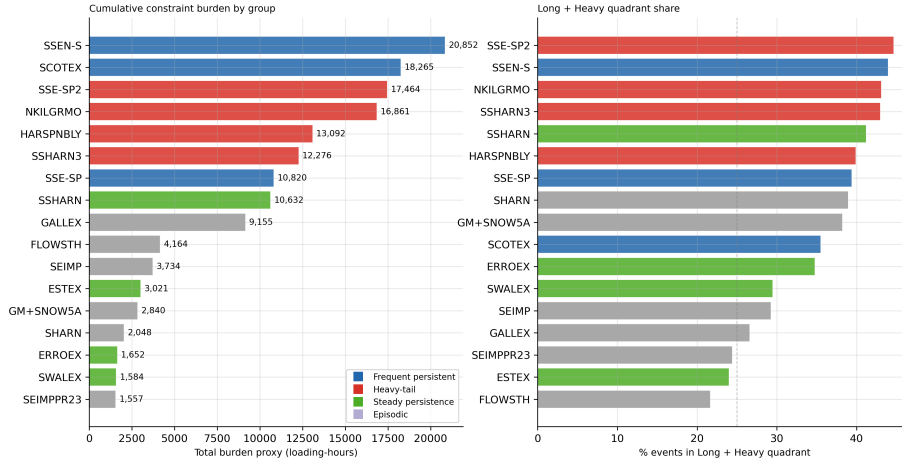


Figure 2 — Constraint Burden Concentration. Operational burden is hyper-concentrated within a small number of North-West transmission boundaries.

- **The 1.7% System-Level Capture Rate:** A representative 50MW 4h battery captures approximately 1.7% of the total system constraint opportunity. While this is not a meaningful benchmark for individual asset selection (as no single asset can capture the entire North-West), it is a critical insight for system planners: **storage at this scale cannot substitute for strategic transmission reinforcement.**
- **Power Before Energy:** The primary limiting factor is instantaneous power capability (Static Failure), not energy depletion. Most missed opportunity occurs because the asset lacks the MW rating to respond to peak constraint events.

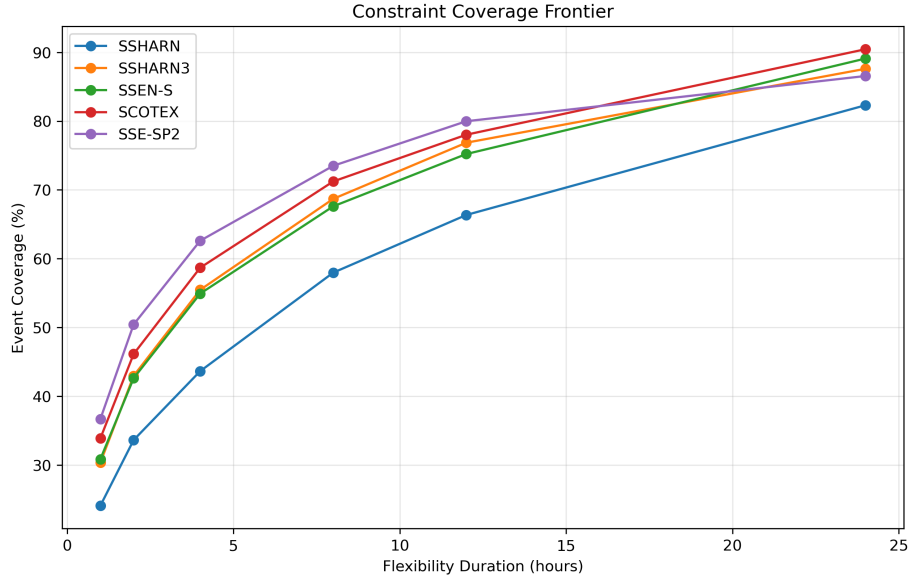
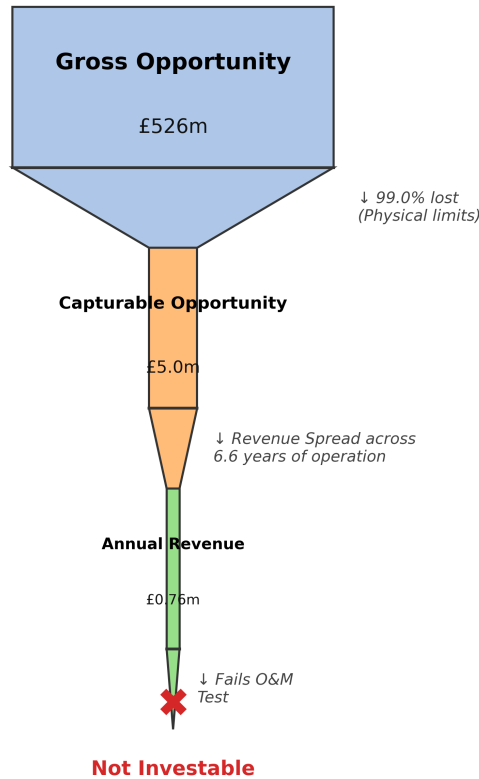


Figure 3 — *The Coverage Frontier. Increasing instantaneous power capability captures materially more value than increasing energy duration alone.*

2. The Economic Reality: The Merchant Case is Structurally Impaired

NB15 introduced rigorous, pre-tax, unlevered project finance modelling to test standalone viability. The results demonstrate that constraint revenue acts only as a marginal top-up to a broader, highly pressured revenue stack.

The Constraint Opportunity Funnel From Theoretical System Value to Investment Reality



Key Insight: Constraint-management revenue alone does not support standalone battery investment.

Figure 4 — The Opportunity Funnel. Theoretical system value is filtered by physical friction and time, collapsing to a fraction of its original scale before becoming investable revenue.

- **Structural Cash Deficit:** Across 40 tested configurations and 5 constraint groups, **100% of assets failed the Revenue Sufficiency Test**. The best-case scenario generated enough constraint revenue to cover only **5%** of annual Fixed O&M.
- **Zero Probability of Success:** A 10,000-iteration Monte Carlo simulation, stress-testing wide distributions of revenue, Capex, Opex, and discount rates, resulted in a **0.0000% probability of positive NPV** for all configurations when relying on constraint revenue as a primary driver.
- **The Viability Threshold:** To achieve a break-even NPV, a trading desk must generate between **£100,000 and £380,000 per MW per year**

from stacked markets (Arbitrage, Capacity Market, FFR), depending on asset duration.

3. The Strategic Tension: A Market Trapped in a Pincer Movement

The most critical finding of this project is the divergence between what the grid physically needs and what the market will fund.

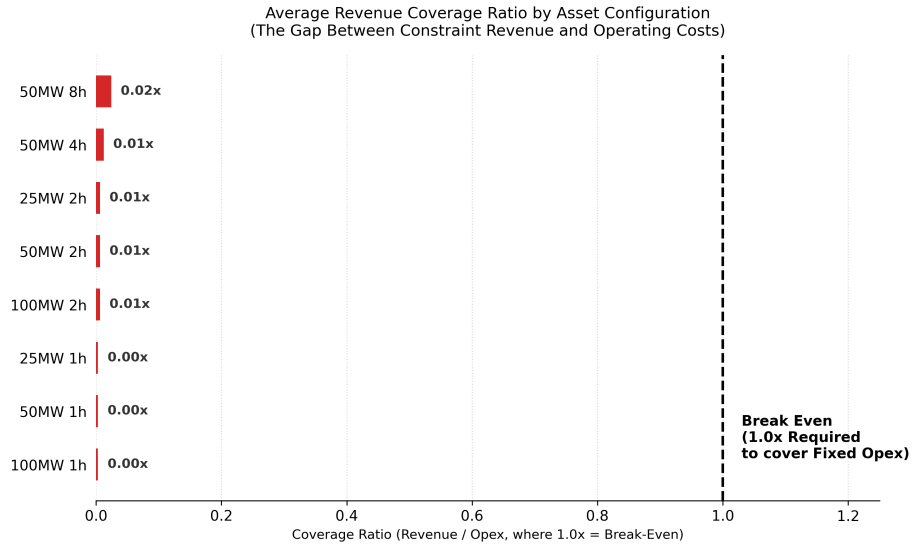


Figure 5 — Revenue Sufficiency Test. Constraint-management revenue fails to cover basic operating costs for all tested configurations.

- **The Long End (4h–8h) has a Relative Physical Advantage, but is Unfundable:** While longer-duration assets capture more constraint value (8h > 4h > 2h), the absolute revenue remains modest across all configurations (£18k–£73k/year). The massive energy Capex of a 4h+ asset requires a trading performance of >£220,000/MW-year to break even. This “Physical Sweet Spot” is a relative, not absolute, statement: longer duration is physically preferable, but the absolute revenue is not economically sufficient.
- **The Short End Faces Growing Revenue Compression:** The market has historically funded 1h–2h assets because their lower Capex requires only £100,000–£140,000/MW-year to break even. However, the connection queue is now oversaturated with ~1.6-hour assets targeting short-duration frequency response (DC/DM) and price surges. This massive oversupply is crashing ancillary service prices, driving the merchant revenue stack below the break-even threshold.

Conclusion: The market is building short-duration assets for financial survival, even though these assets are physically misaligned with the long-duration nature of the underlying transmission bottleneck. Furthermore, the primary business case for those short-duration assets (merchant surges) is collapsing due to oversupply.

4. Strategic Implications

For BESS Developers & Investors

1. **Constraint Revenue Cannot Rescue a Broken Merchant Case:** Do not underwrite projects based on constraint-relief revenue. It must be modelled as a marginal top-up (covering <5% of Opex). If the underlying merchant arbitrage/ancillary case is impaired by cannibalisation, constraint revenue will not save it.
2. **Cost Deflation Alone Cannot Bridge the Gap:** Even under aggressive Capex assumptions (£200/kWh), long-duration assets (4h+) require stacked revenue targets (>£160k/MW-year) that are not currently achievable in GB merchant markets.
3. **Scope Caveat — Co-location and Hybridisation:** This analysis assumes a standalone merchant BESS. Co-location with renewables (sharing grid connection costs) or hybridisation may improve project economics. However, the fundamental tension—that constraint relief is a marginal top-up, not a primary business case—remains relevant for all BESS configurations.

For NESO & Network Planners

1. **Wires are Mandatory; Batteries are Buckets:** A single BESS asset captures <2% of the total system constraint opportunity. Storage can alleviate marginal operational stress, but it cannot substitute for strategic transmission reinforcement (e.g., Eastern Green Link) when congestion becomes structural.
2. **The market shows limited evidence of self-correcting under current incentive structures:** Because the short-duration merchant market is broken and the long-duration market is unfundable, NESO cannot rely on merchant developers to organically deploy the 4h–8h assets required to manage the transition before wires are finished.

For Ofgem & Policymakers

1. **Proven Market Failure Requires Structural Intervention:** The market failure is not simply that revenues are low — it is that the **price signal for duration is structurally weak**. Short-duration assets compete on a crowded field for transient events; long-duration assets provide

sustained relief but are not compensated for that persistence. The Transitional Duration Capacity Market (TDCM) and Locational Energy Pricing (LEP) are therefore not discretionary policy options — they are structural necessities to align financial incentives with physical reality.

5. Conclusion

The North-West transmission system exhibits persistent, clustered congestion that cannot be fully explained by short-term market volatility.

While future market conditions may evolve, the magnitude of the gap between physical need and economic viability (a factor of $\sim 5\text{--}10\times$ in required revenue) suggests that incremental price volatility alone is unlikely to resolve the structural tension. A fundamental market design intervention — not passive price evolution — is required.

Future system resilience will depend not on hoping merchant batteries will organically solve grid bottlenecks, but on the successful, coordinated evolution of **strategic transmission reinforcement** (the physical pipe) and **targeted market design reform** (aligning financial incentives with physical reality).

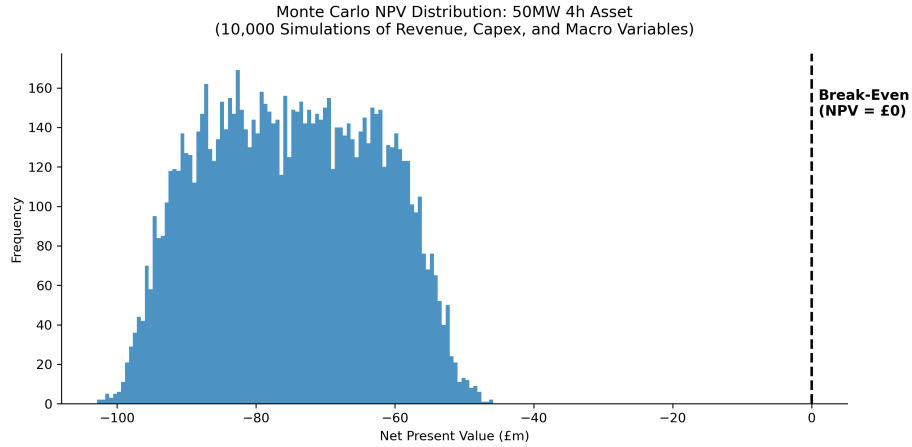


Figure 6 — Probabilistic Reality Check. Across 10,000 simulations of market and cost variables, the probability of a positive NPV remains exactly 0.00%.

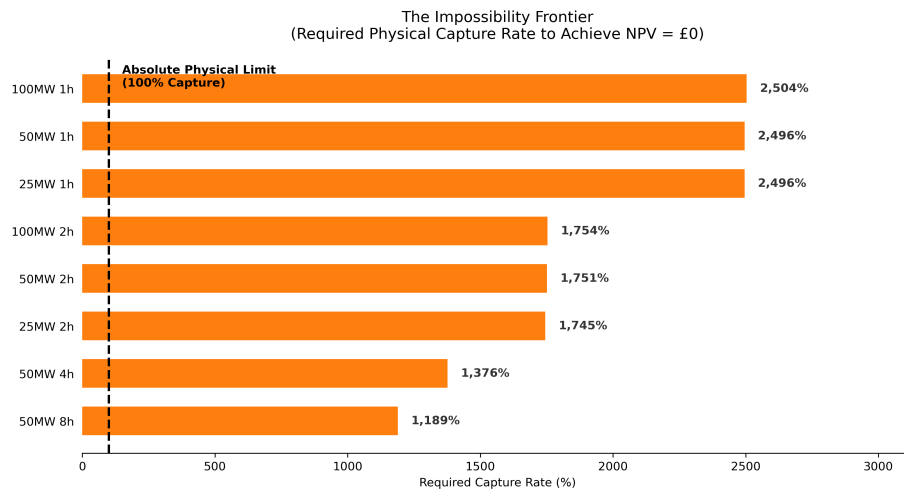


Figure 7 — The Impossibility Frontier. To achieve break-even on constraint revenue alone, assets would need to capture >100% of the physical opportunity, violating the laws of physics.

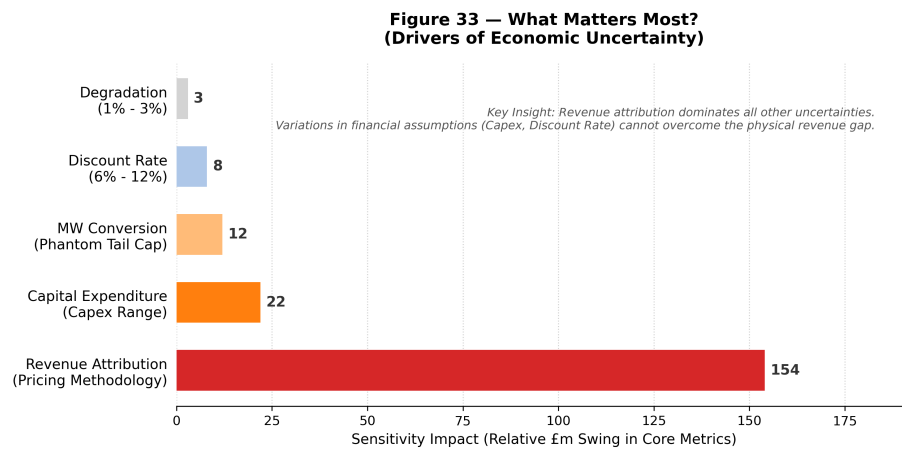


Figure 8 — Sensitivity Hierarchy. Revenue attribution and physical limits dominate all financial assumptions (Capex, Discount Rate), proving the robustness of the non-viability conclusion.